

1. Start with This Decision Framework (MOST IMPORTANT)

Before choosing any metric, ask:

Step 1: Is data balanced?

- Balanced → Accuracy, ROC-AUC OK
- Imbalanced → Avoid Accuracy → Use Recall / Precision / F1 / PR-AUC

Step 2: What is more costly?

- False Positive (FP) costly → Precision
- False Negative (FN) costly → Recall
- Both important → F1

Step 3: Do you care about:

- Threshold-based classification → F1, Precision, Recall
- Ranking / probability quality → ROC-AUC, PR-AUC



2. Metric Selection Cheat Sheet (USE THIS IN INTERVIEWS)

Scenario	Metric to Use	Why
Balanced dataset	Accuracy / ROC-AUC	All classes equally important
Highly imbalanced	F1 / AUC	Accuracy misleading
FP is costly	Precision	Avoid wrong positives
FN is costly	Recall	Capture all positives
Both FP & FN costly	F1 Score	Balance both
Need ranking	ROC-AUC	Measures separability
Focus on positive class (rare)	AUC	Better than ROC in imbalance

3. How to Evaluate Metrics (WITH REAL EXAMPLES)

Example 1: Disease Prediction

- FN = very dangerous (missing patient)

👉 Use:

- Recall (primary)
- F1 (secondary)

Interpretation:

- Recall = 0.95 → Good (captures most patients)
- Precision = 0.60 → Acceptable (some false alarms OK)

✗ Don't use Accuracy:

- Model can be 98% accurate but useless
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Example 2: Fraud Detection

- Fraud = rare (1–2%)
- FP → annoying but OK
- FN → huge loss

👉 Use:

- Recall (primary)
- PR-AUC (important)

Why:

- PR-AUC focuses on positive class (fraud)

Interpretation:

- PR-AUC ↑ → model improves fraud detection
- ROC-AUC may look high but misleading

Example 3: Spam Detection

- FP = bad (important emails go to spam)

👉 Use:

- Precision (primary)
- F1 (secondary)

Interpretation:

- Precision = 0.98 → Excellent (almost no important mail lost)
- Recall = 0.70 → acceptable

Example 4: Recommendation / Ads Click Prediction

- You care about ranking probability

👉 Use:

- ROC-AUC
- Log Loss

Interpretation:

- ROC-AUC = 0.90 → strong ranking ability
- Log Loss ↓ → better probability calibration

Example 5: Loan Approval Model

- Both FP (bad loan) & FN (lost customer) matter

👉 Use:

- F1 Score
- ROC-AUC

Interpretation:

- $F1 = 0.82 \rightarrow$ balanced model
 - $ROC-AUC = 0.88 \rightarrow$ good separability
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4. Metric Trade-Off Evaluation (CRITICAL)

Case 1:

Model	Precision	Recall
A	0.90	0.40
B	0.75	0.75

👉 Choose:

- Model A $\rightarrow$ if FP costly
 - Model B $\rightarrow$ if balance needed (better F1)
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Case 2:

Model	Accuracy	F1
A	95%	0.30
B	90%	0.75

👉 Choose Model B

Reason: Accuracy misleading in imbalance

Case 3:

Model	ROC-AUC	PR-AUC
A	0.95	0.60
B	0.92	0.80

👉 Choose Model B (better for rare positives)



5. Threshold Tuning Insight (VERY IMPORTANT)

Metrics change with threshold:

- Lower threshold → ↑ Recall, ↓ Precision
- Higher threshold → ↑ Precision, ↓ Recall

👉 Use:

- Precision-Recall Curve → choose business-optimal point
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6. Multi-Class Evaluation Strategy

Use:

- Macro avg → equal importance to classes
- Weighted avg → respects imbalance
- Micro avg → overall performance

👉 Example:

- Medical multi-class → Macro
 - Production logs (skewed) → Weighted
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7. Common Mistakes (INTERVIEW GOLD)

1. Using Accuracy in imbalanced data ✗
 2. Using ROC-AUC when positive class is rare ✗
 3. Ignoring threshold tuning ✗
 4. Looking at single metric only ✗
 5. Not aligning metric with business goal ✗
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8. Final Practical Rule (REMEMBER THIS)

👉 **Metric = Business Objective Translation**

Business Need	Metric
Catch all positives	Recall
Avoid false alarms	Precision
Balance both	F1
Rank predictions	ROC-AUC
Rare event detection	PR-AUC